

Software Project Management

4th Edition



Chapter 1

An Introduction

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Outline of talk

In this introduction the main questions to be addressed will be:

- What is software project management? Is it really different from 'ordinary' project management?
- How do you know when a project has been successful? For example, do the expectations of the customer/client match those of the developers?

What is a project?

Some dictionary definitions:

"A specific plan or design"

"A planned undertaking"

"A large undertaking e.g. a public works scheme"

Longmans dictionary

Key points above are ***planning*** and ***size***
of task

Jobs versus projects

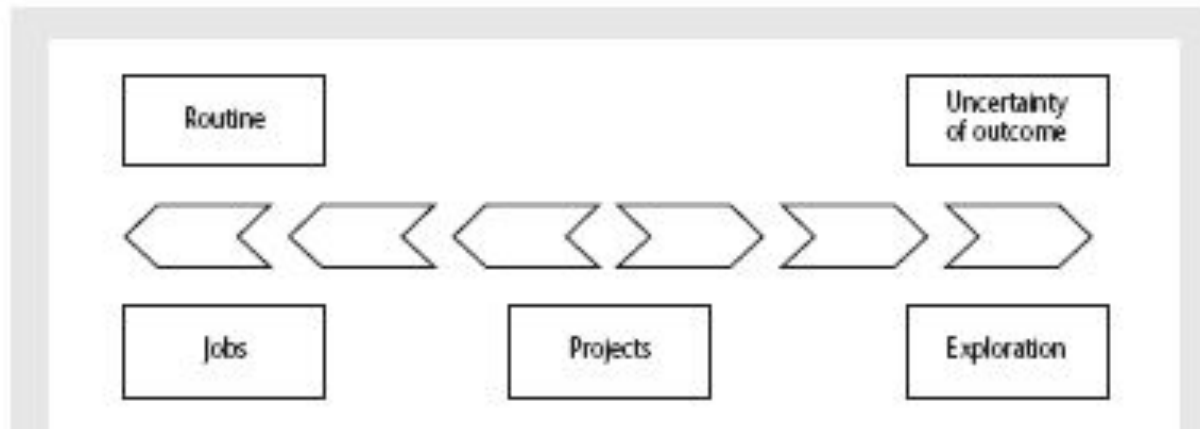


Figure 1.1 Activities most likely to benefit from project management

‘Jobs’ – repetition of very well-defined and well understood tasks with very little uncertainty

‘Exploration’ – e.g. finding a cure for cancer: the outcome is very uncertain

‘Projects’ – in the middle!

Characteristics of projects

A task is more 'project-like' if it is:

- Non-routine
- Planned
- Aiming at a specific target
- Work carried out for a customer
- Involving several specialisms
- Made up of several different phases
- Constrained by time and resources
- Large and/or complex

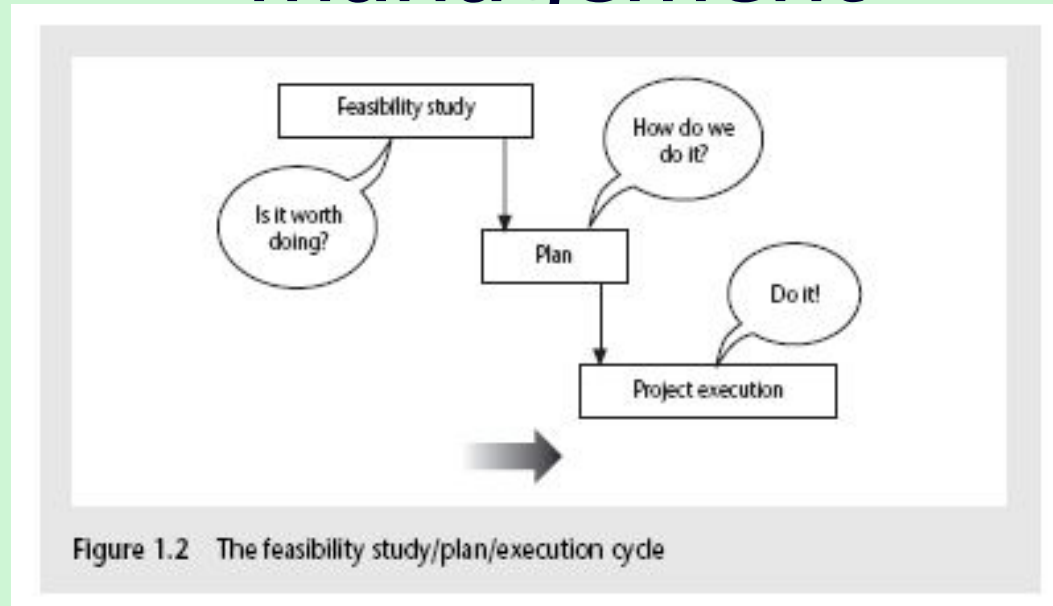
Are *software* projects really different from other projects?

Not really! ...but...

- Invisibility
- Complexity
- Conformity
- Flexibility

make software more problematic to build than other engineered artefacts.

Activities covered by project management



Feasibility study

Is project technically feasible and worthwhile from a business point of view?

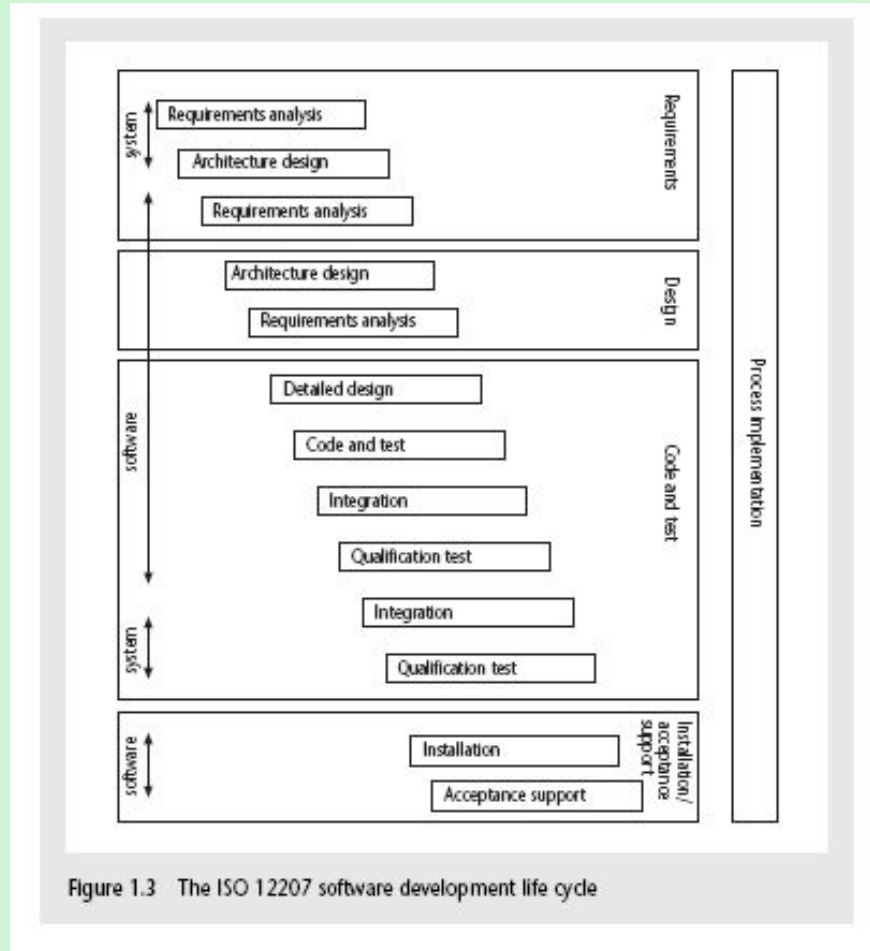
Planning

Only done if project is feasible

Execution

Implement plan, but plan may be changed as we go along

The software development life-cycle (ISO 12207)



ISO 12207 life-cycle

Requirements analysis

- Requirements elicitation: what does the client need?
- Analysis: converting 'customer-facing' requirements into equivalents that developers can understand
- Requirements will cover
 - Functions
 - Quality
 - Resource constraints i.e. costs

ISO 12207 life-cycle

- Architecture design
 - Based on *system requirements*
 - Defines components of system: hardware, software, organizational
 - *Software requirements* will come out of this
- Code and test
 - Of individual components
- Integration
 - Putting the components together

ISO12207 continued

- Qualification testing
 - Testing the *system* (not just the *software*)
- Installation
 - The process of making the system operational
 - Includes setting up standing data, setting system parameters, installing on operational hardware platforms, user training etc
- Acceptance support
 - Including maintenance and enhancement

Some ways of categorizing projects

Distinguishing different types of project is important as different types of task need different project approaches e.g.

- Information systems versus embedded systems
- Objective-based versus product-based

What is management?

This involves the following activities:

- Planning – deciding what is to be done
 - Organizing – making arrangements
 - Staffing – selecting the right people for the job
 - Directing – giving instructions
- continued...

What is management?

(continued)

- Monitoring – checking on progress
- Controlling – taking action to remedy hold-ups
- Innovating – coming up with solutions when problems emerge
- Representing – liaising with clients, users, developers and other stakeholders

Setting objectives

- Answering the question '*What do we have to do to have a success?*'
- Need for a *project authority*
 - Sets the project scope
 - Allocates/approves costs
- Could be one person - or a group
 - Project Board
 - Project Management Board
 - Steering committee

Objectives

Informally, the objective of a project can be defined by completing the statement:

The project will be regarded as a success if.....

Rather like *post-conditions* for the project

Focus on *what* will be put in place, rather than *how* activities will be carried out

Objectives should be SMART

- S** – specific, that is, concrete and well-defined
- M** – measurable, that is, satisfaction of the objective can be objectively judged
- A** – achievable, that is, it is within the power of the individual or group concerned to meet the target
- R** – relevant, the objective must be relevant to the true purpose of the project
- T** – time constrained: there is a defined point in time by which the objective should be achieved

Goals/sub-objectives

These are steps along the way to achieving the objective. Informally, these can be defined by completing the sentence...

**Objective X will be achieved
IF the following goals are all achieved**

A.....

B.....

C..... etc

Goals/sub-objectives continued

Often a goal can be allocated to an individual.
Individual may have the capability of achieving goal, but not the objective on their own e.g.

Objective – user satisfaction with software product

Analyst goal – accurate requirements

Developer goal – software that is reliable

Measures of effectiveness

How do we know that the goal or objective has been achieved?

By a practical test, that can be objectively assessed.

e.g. for user satisfaction with software product:

- Repeat business – they buy further products from us
- Number of complaints – if low etc etc

Stakeholders

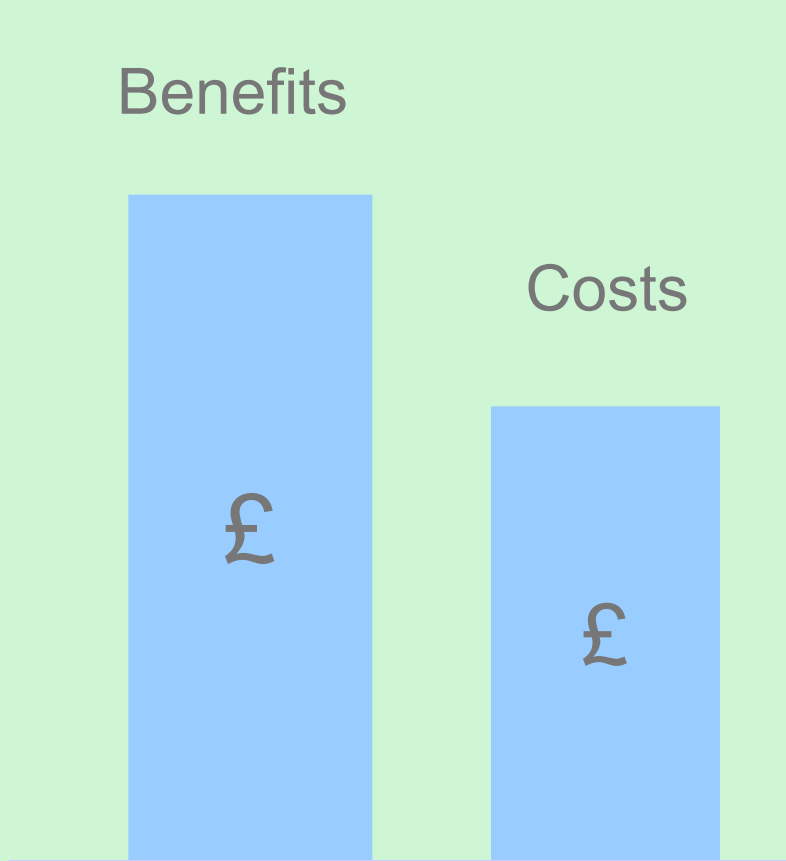
These are people who have a stake or interest in the project

In general, they could be *users/clients* or *developers/implementers*

They could be:

- Within the project team
- Outside the project team, but within the same organization
- Outside both the project team and the organization

The business case



Benefits of delivered project must outweigh costs

Costs include:

- Development
- Operation

Benefits

- Quantifiable
- Non-quantifiable

Management control

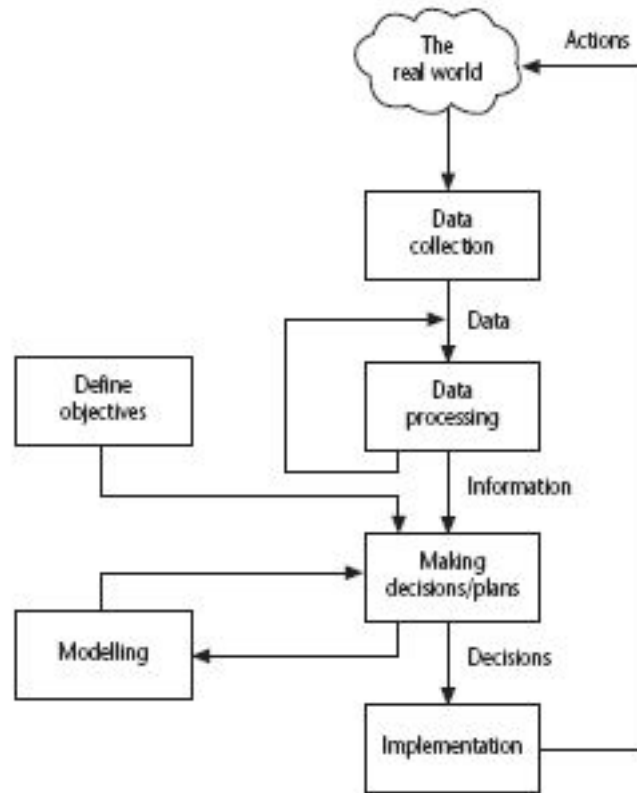


Figure 1.4 The project control cycle.

Management control

Data – the raw details

e.g. '6,000 documents processed at location X'

Information – the data is processed to produce something that is meaningful and useful

e.g. 'productivity is 100 documents a day'

Comparison with objectives/goals

e.g. we will not meet target of processing all documents by 31st March

continued.....

Management control - continued

Modelling – working out the probable outcomes of various decisions

e.g. if we employ two more staff at location X
how quickly can we get the documents
processed?

Implementation – carrying out the remedial actions that have been decided upon

Key points in lecture

- Projects are non-routine - thus uncertain
- The particular problems of projects e.g. lack of visibility
- Clear objectives are essential which can be objectively assessed
- Stuff happens. Not usually possible to keep precisely plan – need for control
- Communicate, communicate, communicate!